

L5



CLASS X - SCIENCE

CONTROL AND COORDINATION

| PRASHANT KIRAD |

LET'S TEST YOUR MEMORY



Q. Which hormone helps regulate blood sugar levels?

- (a) Adrenaline
- ☒ (b) Insulin
- (c) Thyroxin
- (d) Testosterone

LET'S TEST YOUR MEMORY



Q. Which of the following correctly matches the hormone with its gland?

(a) Insulin — Thyroid gland ~~X~~

✓ (b) Adrenaline — Adrenal gland

(c) Thyroxin — Pancreas ~~X~~

(d) Growth hormone — Ovaries ~~X~~

LET'S TEST YOUR MEMORY



Q. How does our body respond when adrenaline is secreted into the bloodstream?

When adrenaline is secreted into the bloodstream:

- ✓ The heart beats faster, increasing the supply of oxygen to the muscles.
- ✓ Blood supply to the digestive system and skin is reduced.
- ✓ More blood is diverted towards the skeletal muscles.
- ✓ The breathing rate increases due to contractions of the diaphragm and rib muscles.

These changes prepare the body to face an emergency situation through the fight-or-flight response.

LET'S TEST YOUR MEMORY



Q. In animals, hormones can be secreted by one organ and can act on multiple organs. Justify this statement by explaining the effect of a single animal hormone on three organs. [CBSE Practice Set-1, 2023]

Adrenaline, secreted by the adrenal glands, is carried through the blood and acts on different organs:

- ✓ 1. Heart: It increases the heartbeat, supplying more oxygen to the muscles.
- ✓ 2. Digestive system and skin: It reduces their blood supply by contracting the muscles around small arteries, diverting blood towards the skeletal muscles.
- ✓ 3. Diaphragm and rib muscles: It increases their contraction, thereby increasing the breathing rate.

Thus, a single hormone can act on multiple organs and coordinate the body's fight-or-flight response

LET'S TEST YOUR MEMORY



Q. A hormone 'X' is secreted into the blood when a person faces a frightening or emergency situation.

(a) Identify hormone 'X' and name the gland that secretes it.

(b) Explain its role in dealing with frightening or emergency situations.

(a) Hormone 'X' is adrenaline, secreted by the adrenal glands.

(b) Adrenaline prepares the body for the fight-or-flight response by:

- Increasing the heartbeat and breathing rate
- Increasing blood supply to the muscles
- Increasing the blood glucose level to provide more energy

These changes help the person respond quickly during an emergency.

LET'S TEST YOUR MEMORY



Q. A doctor has advised Sameer to reduce sugar intake in his diet and do regular exercise after checking his blood test reports.

- 1. Which disease do you think Sameer is suffering from?**
- 2. Name the hormone responsible for this disease and the organ producing the hormone.**

Sameer is most likely suffering from diabetes.

- Hormone responsible: Insulin
- Organ producing it: Pancreas

Insufficient secretion of insulin causes the blood sugar level to rise.



LET'S TEST YOUR MEMORY

Q. In human beings, hormonal action is largely controlled by a mechanism where the secretion of one hormone is regulated by the action of another. An example of blood glucose level control is shown in the diagram below.

(A) What is the mechanism of hormone action known as?

(B) Which is the sensor X that helps in detecting blood glucose level? What would happen if such a mechanism is absent in humans?

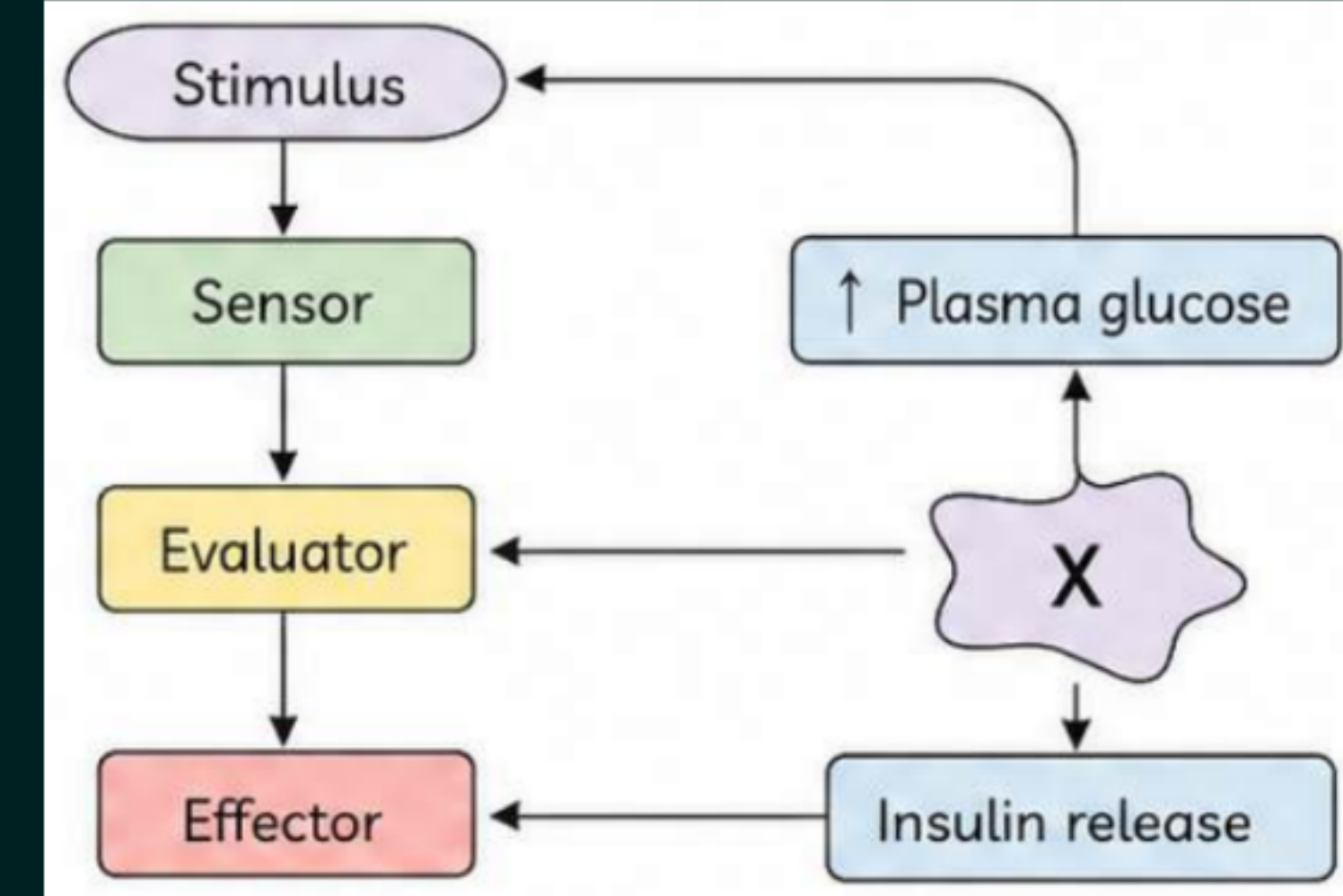
(A) The mechanism is called a feedback mechanism.

(B) X represents the cells of the pancreas, which detect changes in the blood glucose level.

When the blood glucose level rises, the pancreas secretes more insulin. As the blood glucose level falls, insulin secretion is reduced.

If this feedback mechanism were absent, insulin would not be released in the correct amount. The blood sugar level would become uncontrolled, leading to harmful effects such as diabetes.

Increased blood glucose → Pancreas detects it → More insulin is released → Blood glucose decreases → Insulin secretion is reduced

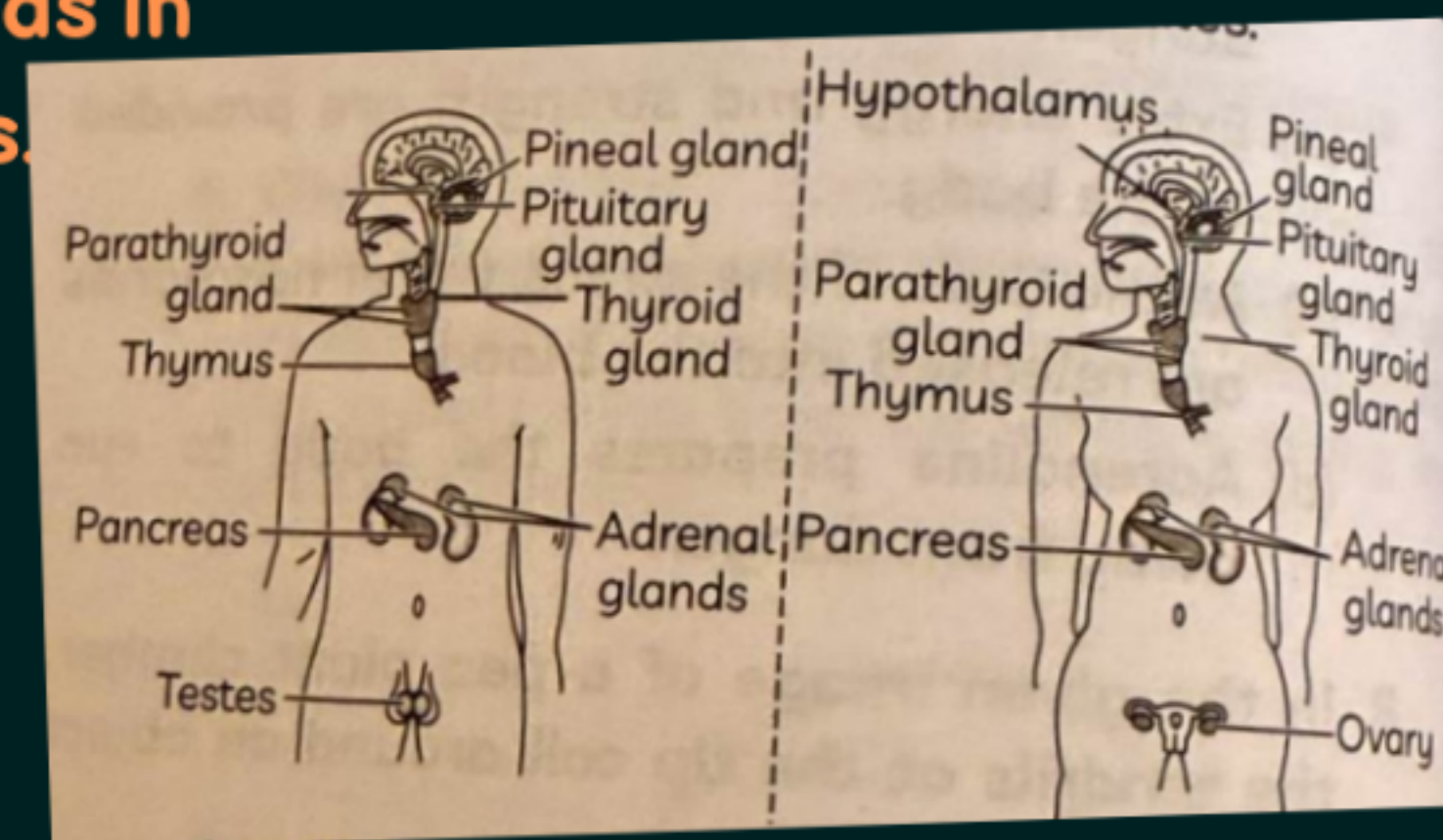


LET'S TEST YOUR MEMORY



Q. Renu was interested in knowing about the various endocrine glands in the human body, the hormones secreted by them and their functions. The given figure shows the endocrine glands in human males and females.

- (A) Which hormone is produced from the pancreas that helps in regulating blood sugar levels?**
- (B) Which endocrine gland is known as the master gland in human beings?**
- (C) Which gland secretes the growth hormone?**



- (A) The pancreas produces insulin, which regulates blood sugar levels.**
- (B) The pituitary gland is known as the master gland.**
- (C) The pituitary gland secretes the growth hormone.**



LET'S TEST YOUR MEMORY

Q. The thyroid hormone regulates metabolism in the body. The thyroid gland primarily releases two hormones: Thyroxine (T_4) and Triiodothyronine (T_3), collectively known as thyroid hormones. The thyroid-stimulating hormone (TSH), produced by the pituitary gland, regulates the release of these hormones.

Ram, a resident of a remote Himalayan village, visited a doctor with complaints of tightness in the throat and a hoarse voice. The doctor advised a thyroid profile and a blood sugar test. Given below are the test results:

Test	Normal/Reference Range	Amount in Sample
TSH	0.4–5 mIU/L	6 mIU/L
T_3	4.6–12 $\mu\text{g/dL}$	1.76 $\mu\text{g/dL}$
T_4	80–180 ng/dL	21 ng/dL
Blood sugar (post-meal)	170–200 mg/dL	180 mg/dL

Based on this information, answer the following questions:

(A) What deficiency disorder might Ram be suffering from? Provide a justification.

(B) Mention two reasons why the doctor recommended a thyroid test.

(C) What role does iodine play in the production of thyroid hormones?

(D) Suggest a dietary recommendation for Ram to help him manage his condition if iodine deficiency is confirmed.

Goiter

(A) Ram may be suffering from goitre due to iodine deficiency, indicated by throat tightness and hoarse voice.

(B) Thyroid test was advised to check:

1. Enlargement of the thyroid gland
2. Abnormal thyroid hormone levels

(C) Iodine is essential for the production of thyroxin.

(D) He should consume iodised salt and iodine-rich food.

Coordination in Plants

Coordination in Plants

- Plants do not have a nervous system or sense organs like eyes, ears, and nose as seen in animals.
- Even then, they can detect and respond to changes in their environment.
- Plants coordinate these responses with the help of chemical substances called hormones instead of nerves.



Q. How is coordination in plants different from animals?

Animals	Plants
Have specialised tissues (<u>nerves</u> , <u>muscles</u>) for conduction of information.	No specialised tissues for <u>conduction of information</u> .
<u>Movement happens due to change in shape of muscle cells</u> (caused by special proteins).	<u>Movement happens due to change in amount of water in cells</u> (swelling/shrinking).

- Both use chemical signals (hormones) for control and coordination.
- In both, cells change shape to bring about response.
- Both help the organism adapt and survive in changing environment.

Response in Plants

✓ Not related to growth

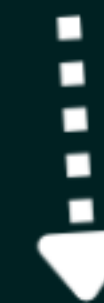
Non-directional response to stimulus

✓✓ Nastic movements

✓ Related to growth

Directional growth towards/away from stimulus

✓ Tropic movements



Tropic movements

→ Tropism

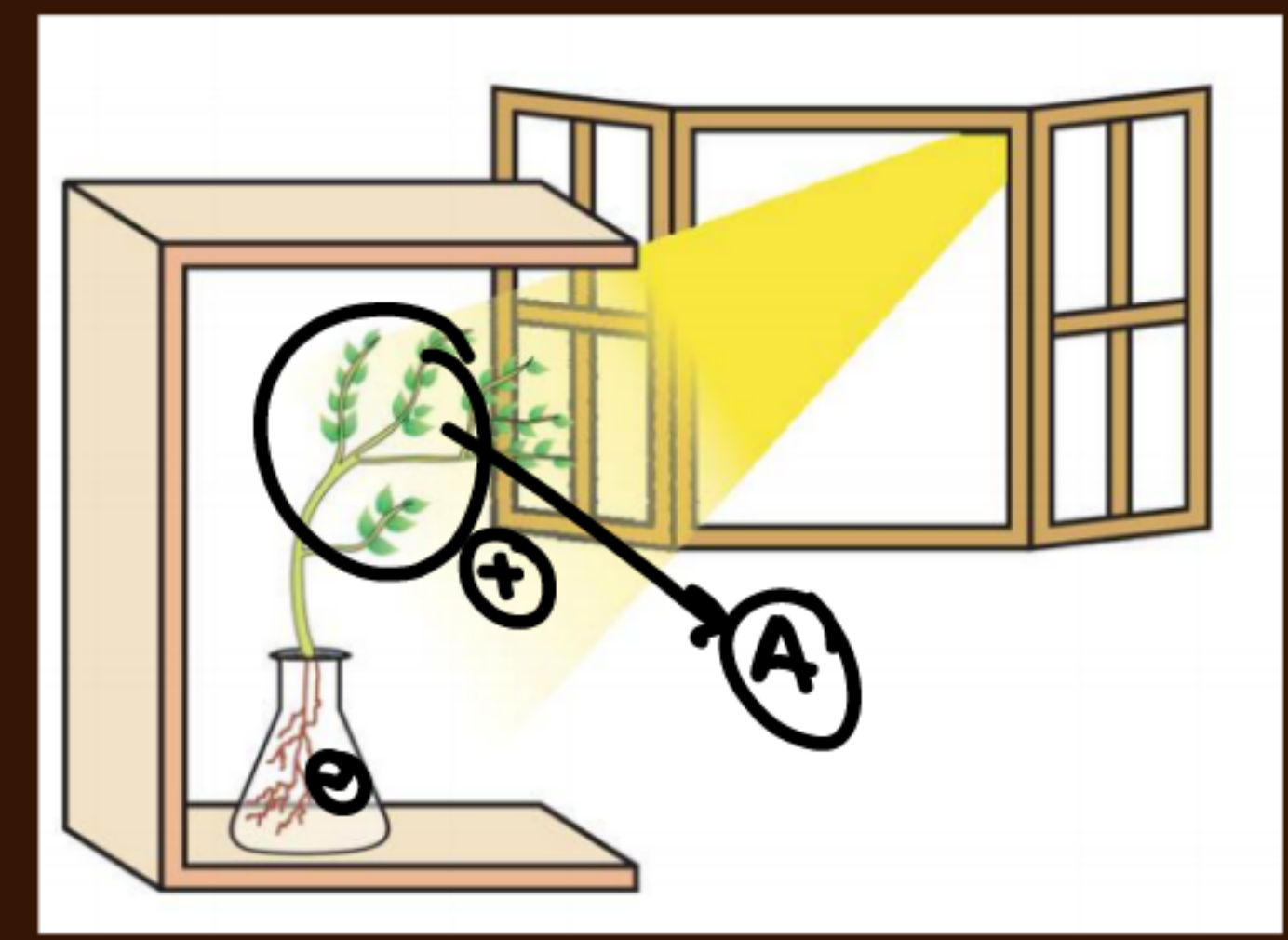
- Tropism is the growth movement of a plant part in response to an external stimulus.
 - The direction of the stimulus determines the direction of plant growth.
 - It is a directional movement caused by growth.
- Growth towards the stimulus is called **positive tropism**.
 - Growth away from the stimulus is called **negative tropism**.

Phototropism

- The movement of a plant part in response to light is called phototropism.
- **Positive Phototropism:** The plant's stem grows toward light.
- **Negative Phototropism:** The plant's roots grow away from light.



Germinated bean seeds kept in a flask facing light show shoots bending towards light and roots bending away from light. When the flask is turned, the new growth changes direction according to light, but the old parts do not. This shows that shoots are positively phototropic and roots are negatively phototropic.

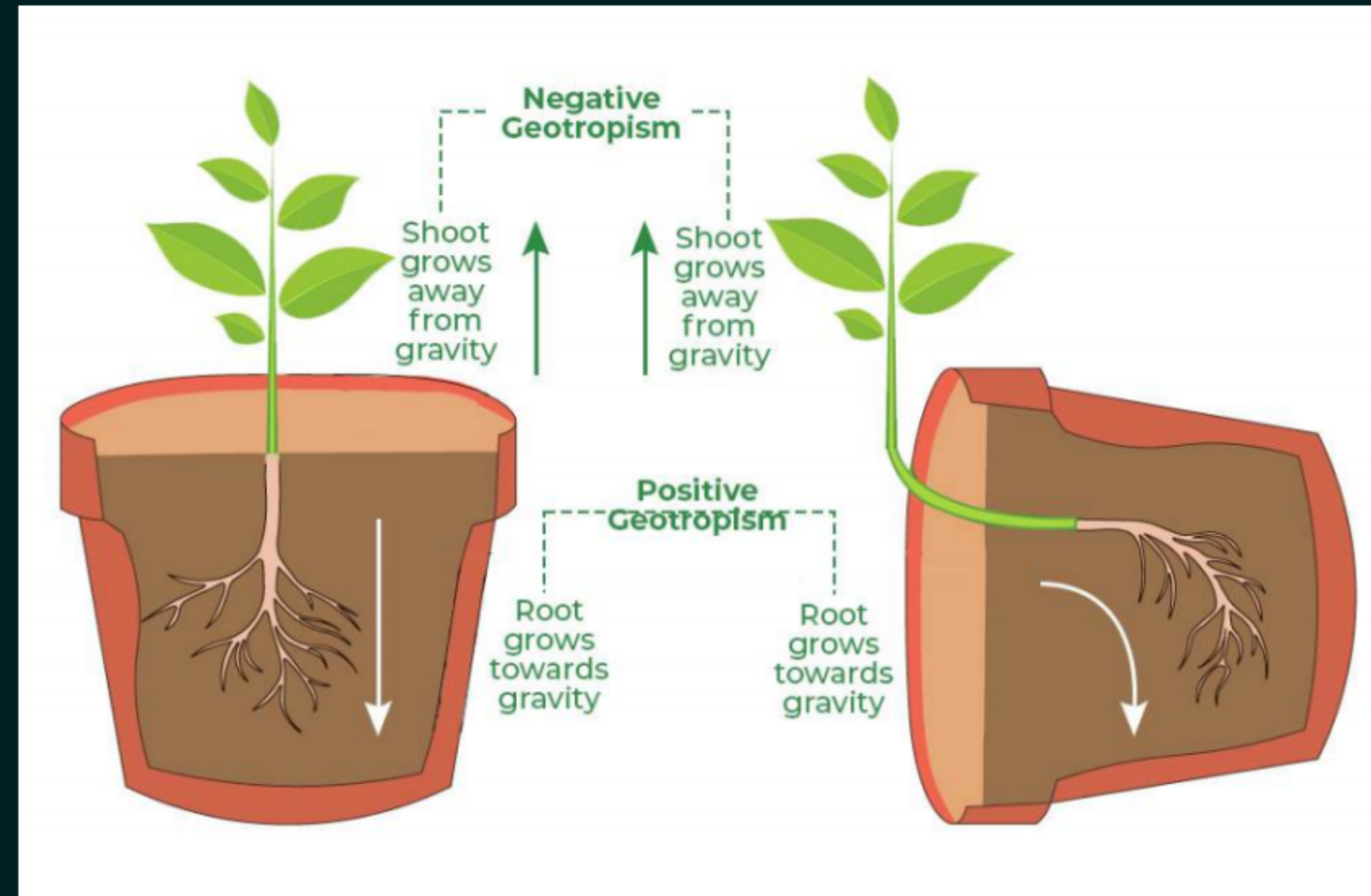


Geotropism

↳ Gravity



- Geotropism is the growth movement of a plant part in response to gravity.
- Roots grow in the direction of gravity and show **positive geotropism**.
- Stems grow against gravity and show **negative geotropism**.

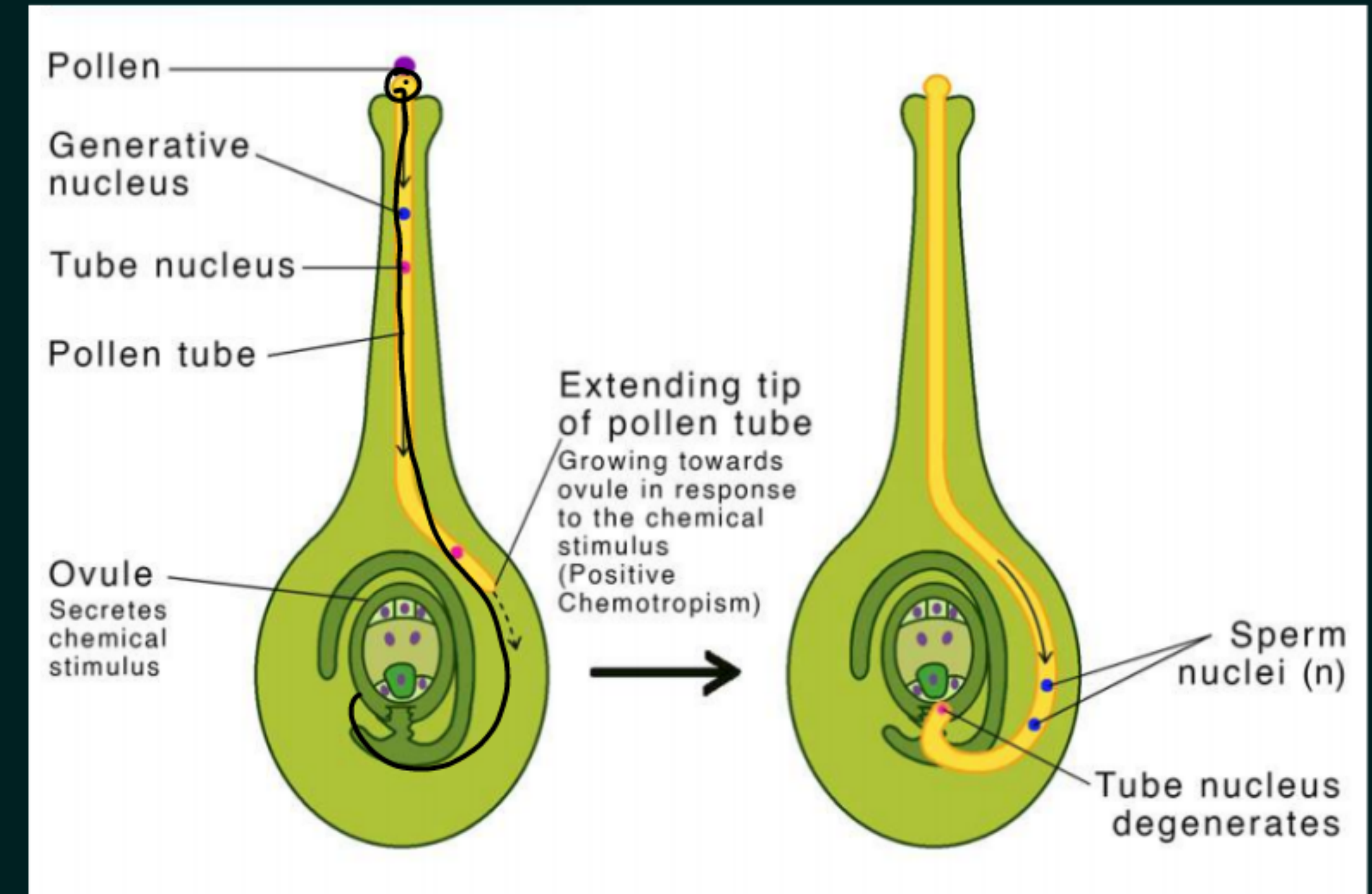


Chemotropism

→ Chemical

- Chemotropism is the growth movement of a plant part in response to a chemical stimulus.
- Growth towards the chemical stimulus is called **positive chemotropism**.
- Growth away from the chemical stimulus is called **negative chemotropism**.

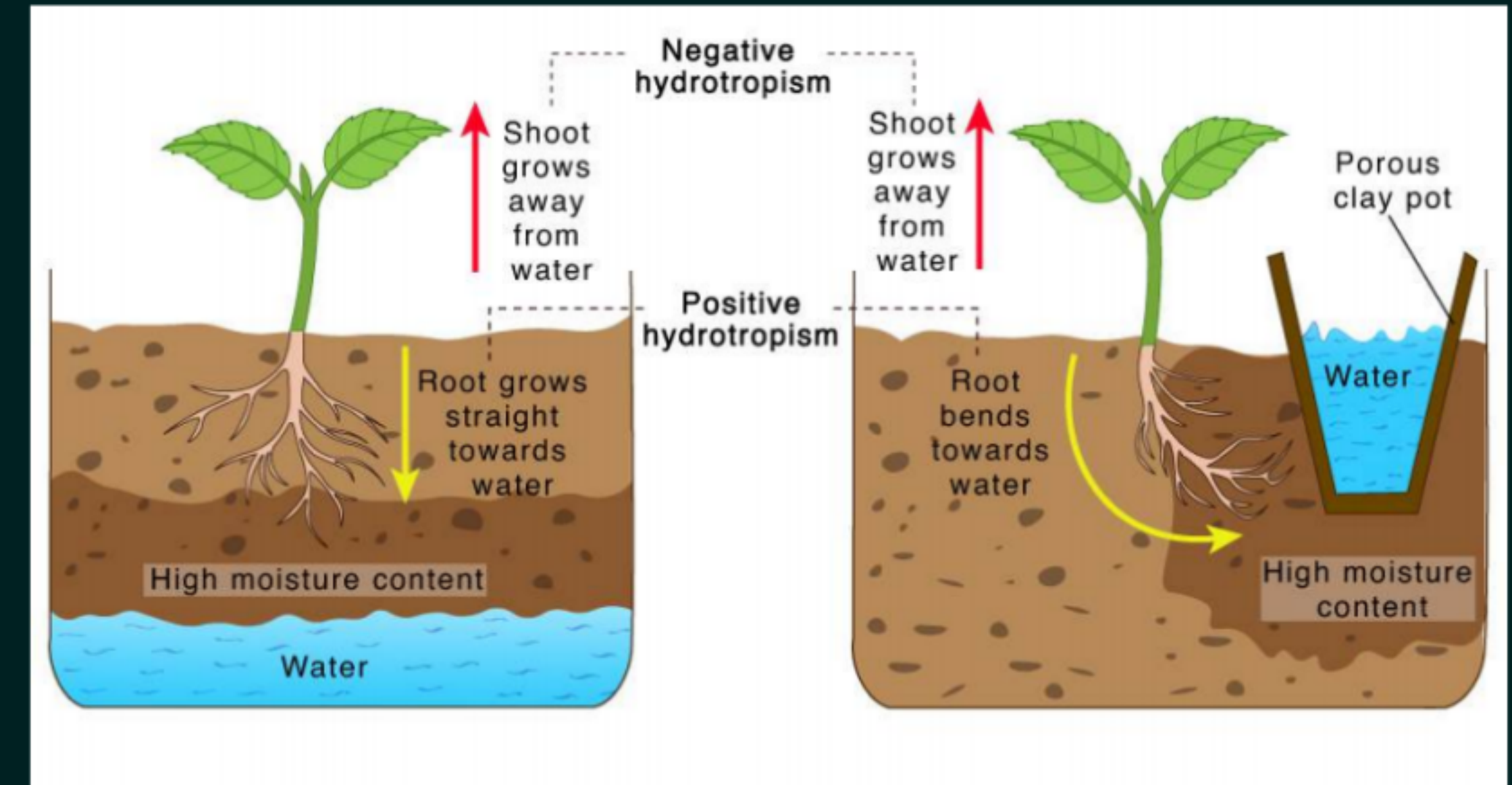
(K) Example: Growth of pollen tubes towards ovules



Hydrotropism

→ water

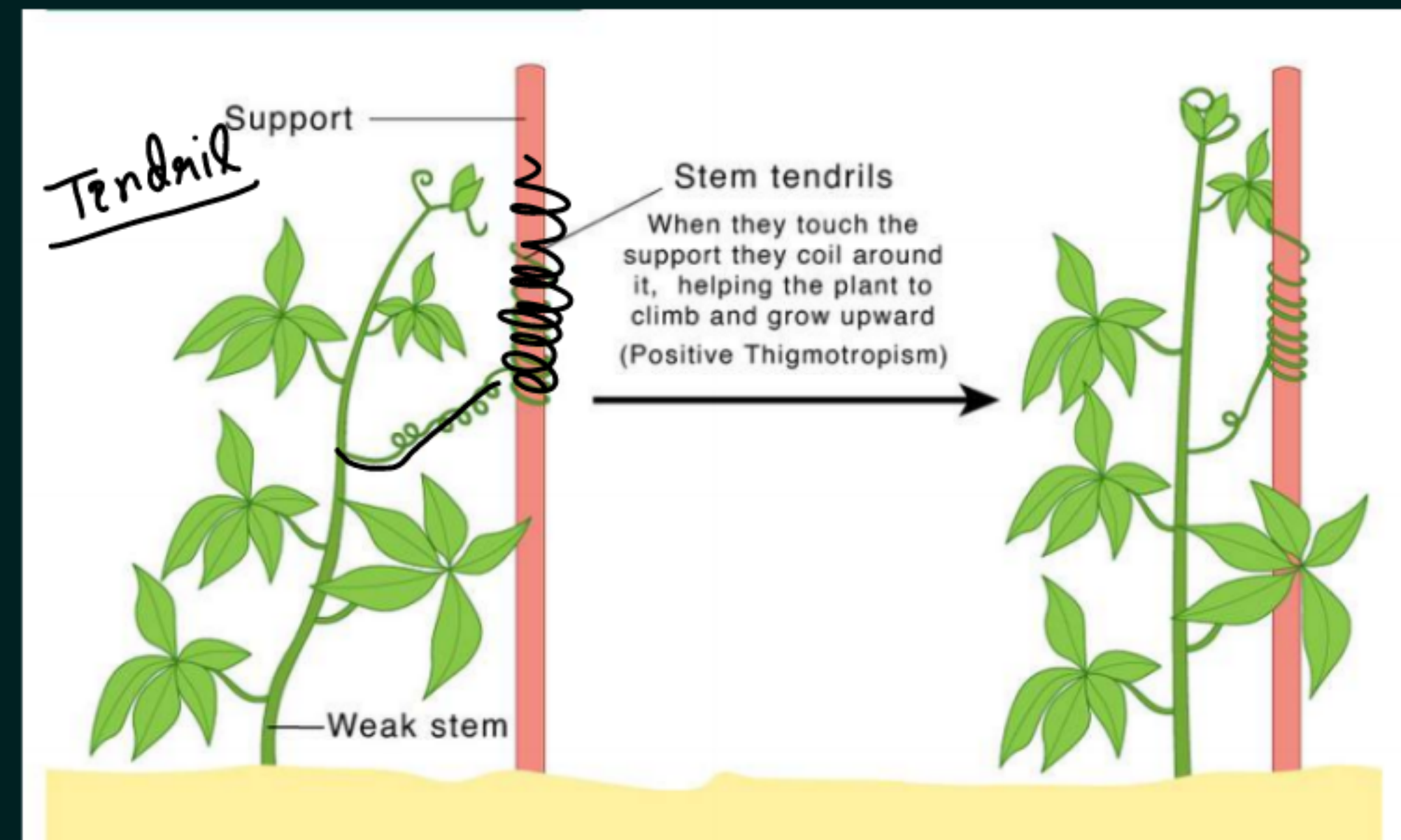
- Hydrotropism is the growth movement of a plant part in response to water.
- Roots usually grow towards water and show positive hydrotropism.
- Growth away from water is called negative hydrotropism



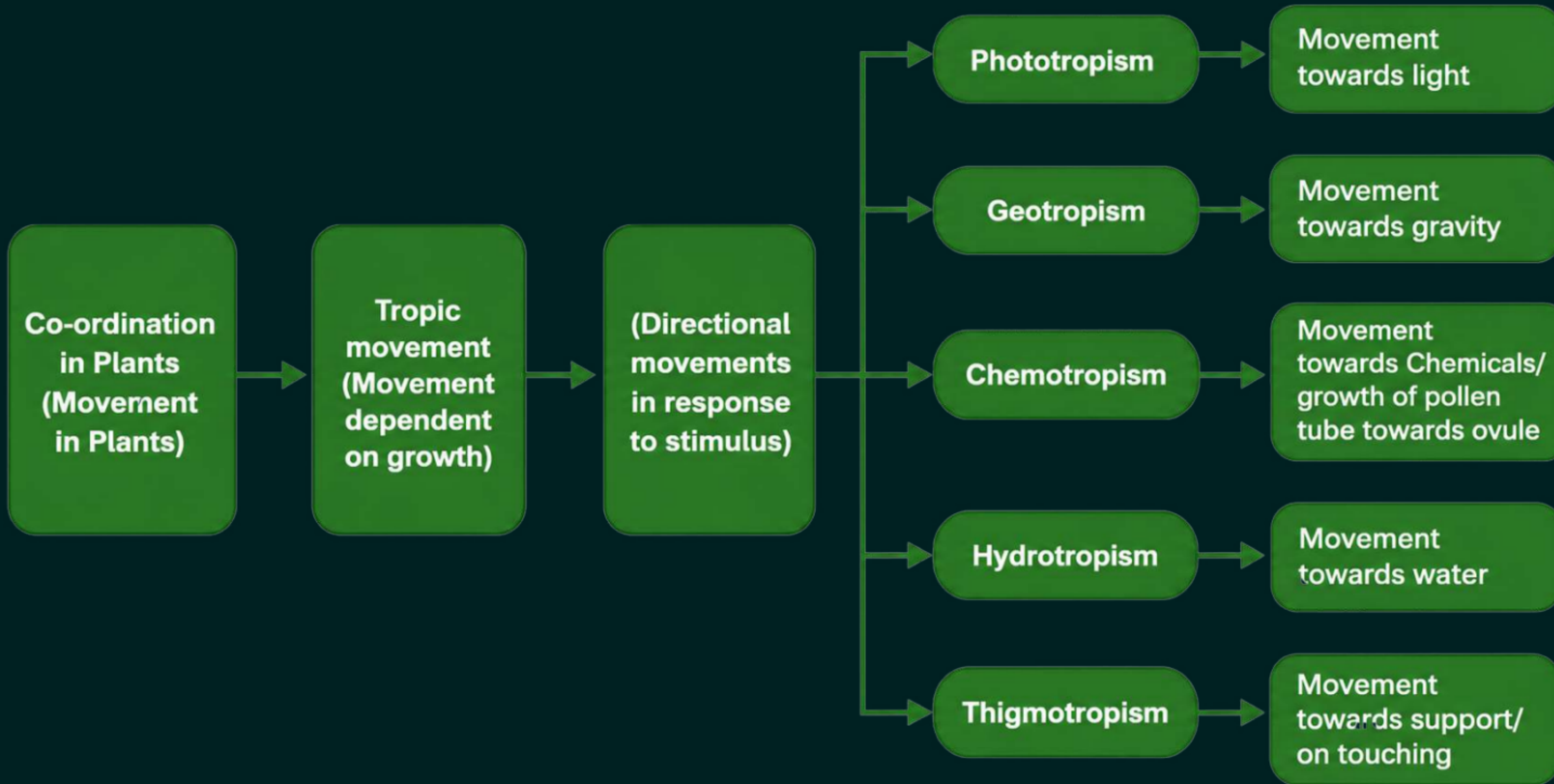
Thigmotropism

→ Touch

- Thigmotropism is the directional growth movement of a plant part in response to touch or physical contact.
- Tendrils of climbing plants grow towards a support when they come in contact with it.
- The tendrils coil around the support to provide stability to the plant.
- Tendrils show positive thigmotropism.



Short summary of tropic movements



Nastic movements

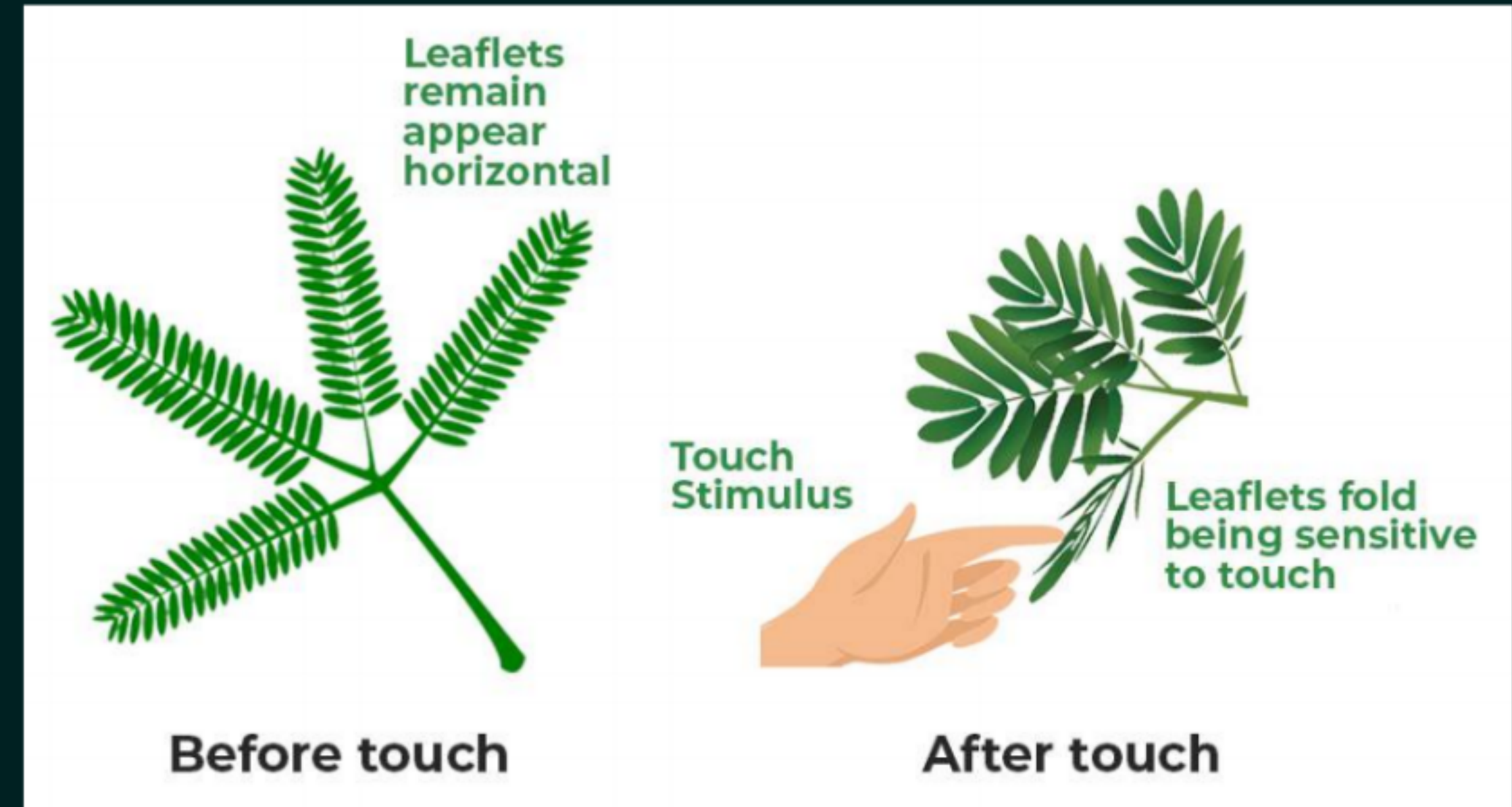
→ Non-growth Related

Thigmonasty :

→ Touch

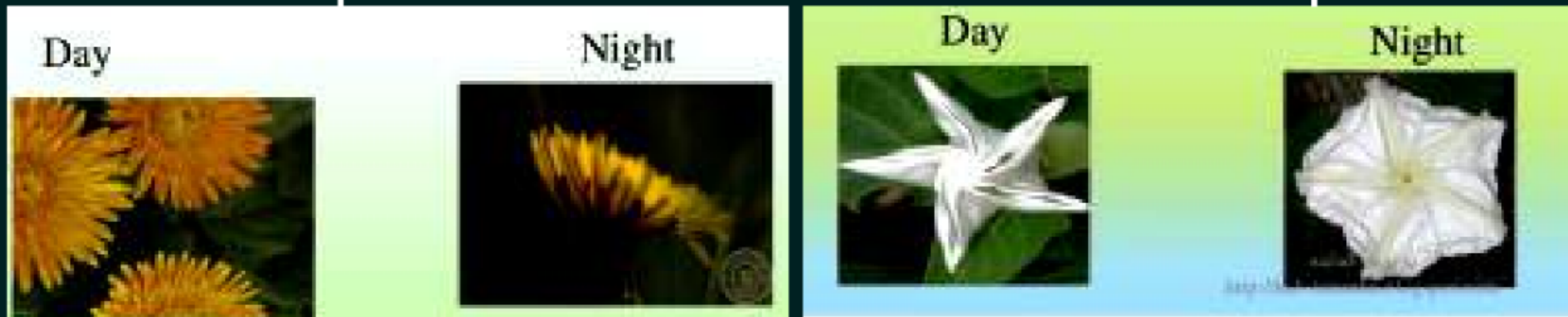
Non-directional movement in response to touch.

In the sensitive plant (Mimosa pudica or chhui-mui), touching the leaflets causes them to fold and droop quickly



Photonasty :

Non-directional movement of flower petals in response to light. Flowers open when the inner petal surfaces grow more and close when the outer surfaces grow more. e.g. Dandelion and Tulip (open in light) and Moonflower or Evening Primrose (open in dark/close in sun).



2043 Mark

(T)

(R)

Tropic Movement	Nastic Movement
<u>Directional movement</u> in response to a stimulus	<u>Non-directional movement</u> in response to a stimulus
<u>Depends on the direction of the stimulus</u>	Does not depend on the direction of the stimulus
<u>Usually related to growth</u>	<u>Usually not related to growth</u>
<u>Movement is generally slow</u>	Movement is usually fast
Caused by unequal growth of plant parts	Caused by changes in water content or turgor pressure
Example: Shoot bending towards light	Example: Folding of Mimosa leaves on touch

bye.

